Lecture 19: **Approximate Bayesian Computation**

October 28, 2026

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prior is chosen, so let's think more about the **likelihood**

likelihood — probability distribution of $\mathscr{D}$ given θ

$$p(\mathscr{D} \mid \theta)$$

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if we sample the likelihood

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common **assumption**: likelihood is Gaussian

often reasonable due to central limit theorem but not always!
(*small sample size, skewed distributions, correlated samples*)

what if we don't have access to the likelihood?

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we often have **simulations** that can approximate the data generation process, can we use them for inference?

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a realization of the **data generation process**

we often have simulations that can approximate the data generation process, can we use them for inference? **simulation-based inference**

also known as "likelihood-free" or "implicit likelihood" inference

approximate bayesian computation (ABC)

ABC — **simulator** $F(\theta)$ instead of the likelihood

$\mathscr{D}' \sim \cancel{p(\mathscr{D} \mid \theta)}$ we don't have access to the explicit likelihood

$\mathscr{D}' \sim F(\theta)$ but we have a simulator F takes θ as input and simulates the data generation process

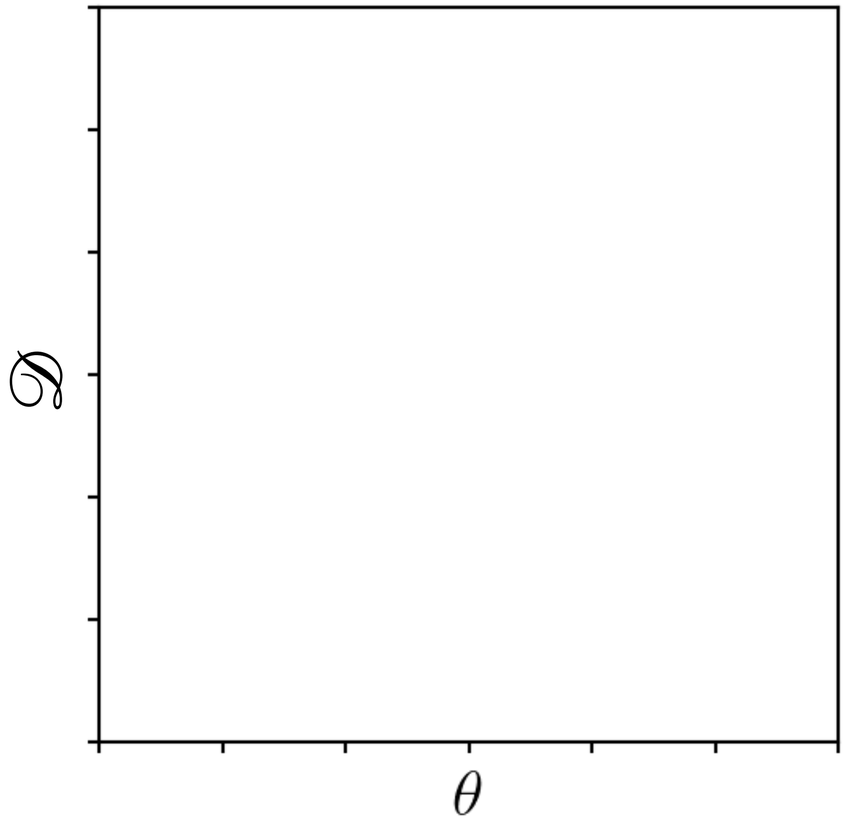
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including noise, systematics, etc

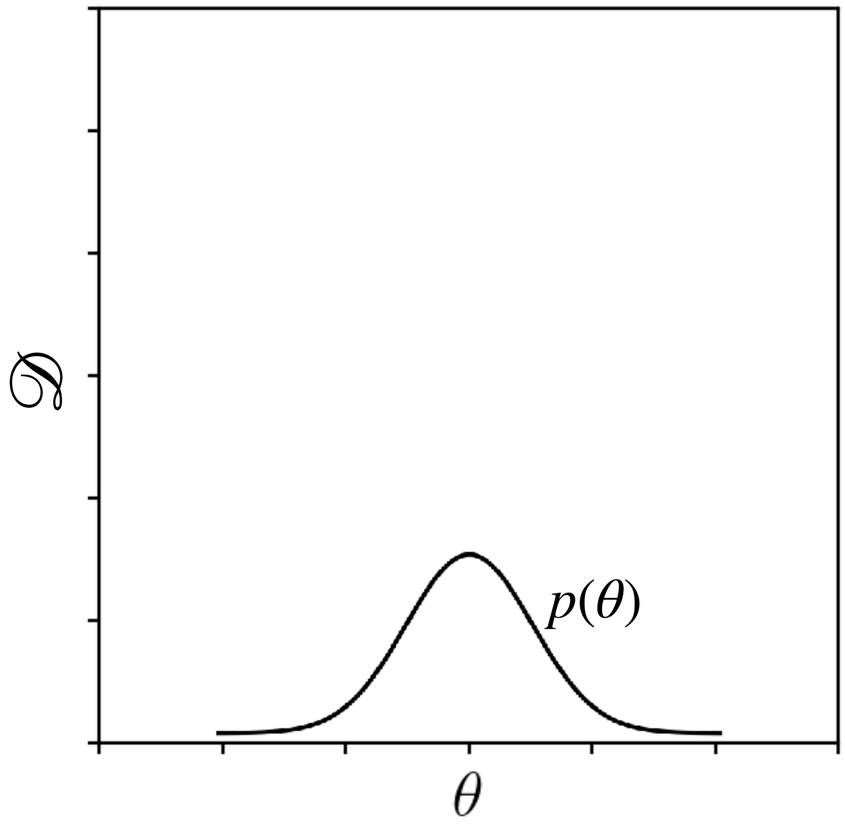
ABC in 1-dimensions: θ and $\mathscr{D}$ are both 1D



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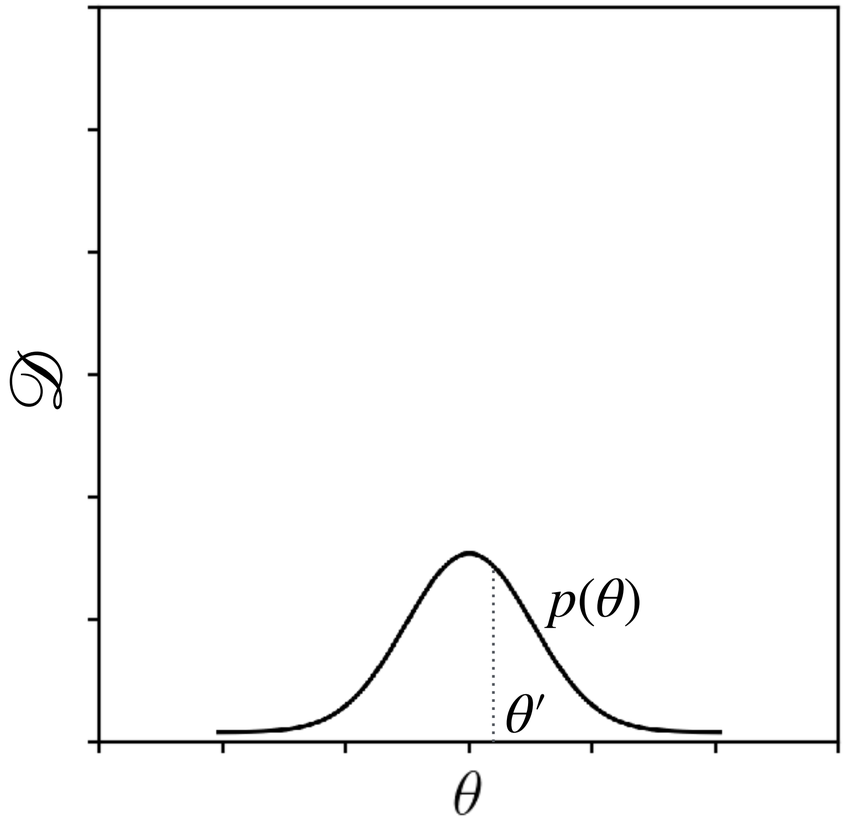
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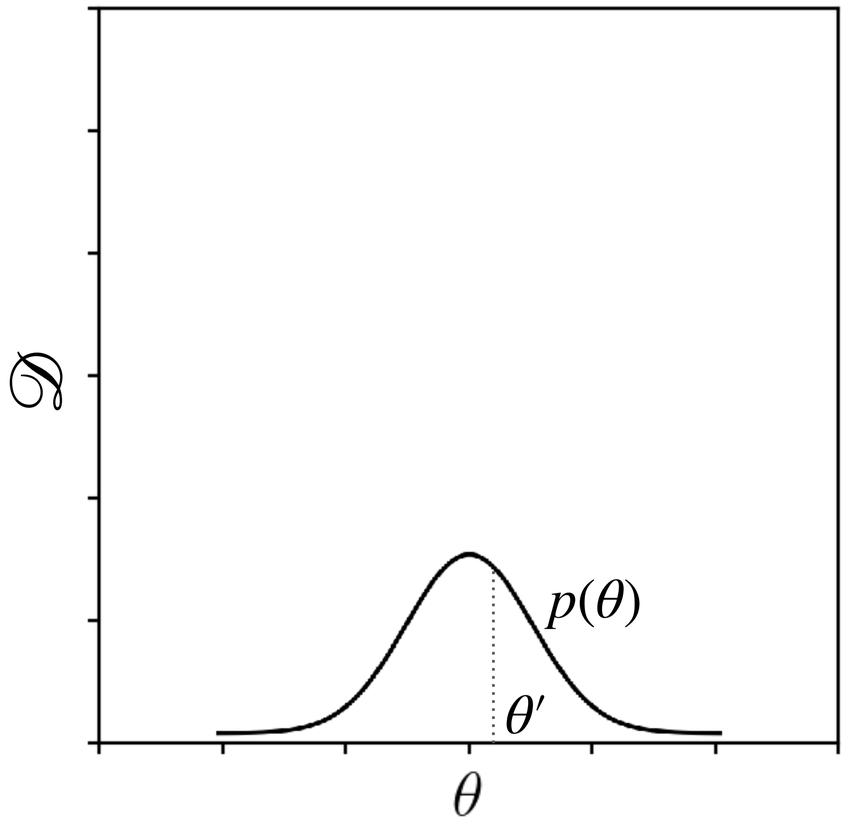
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1. sample the prior

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2. run simulator

$$\mathscr{D}' \sim F(\theta)$$



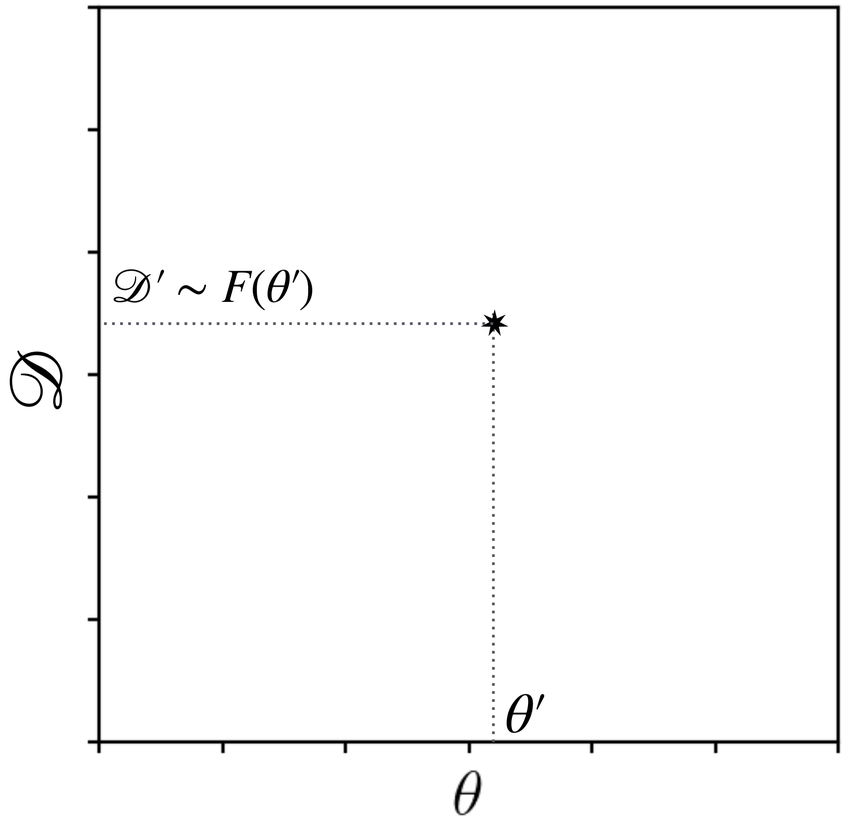
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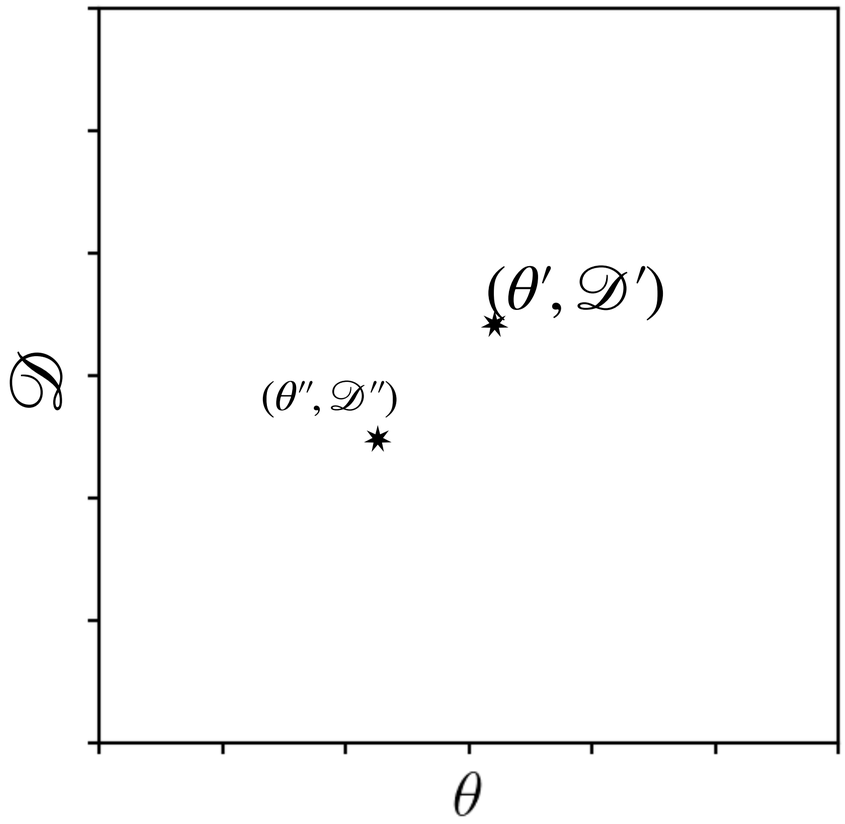
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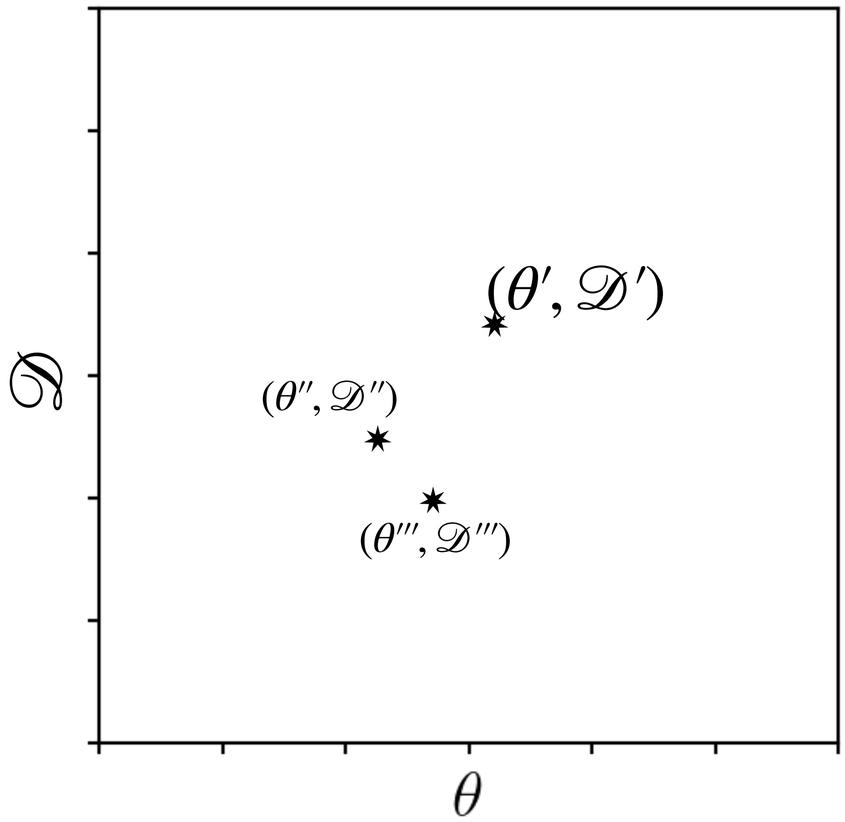
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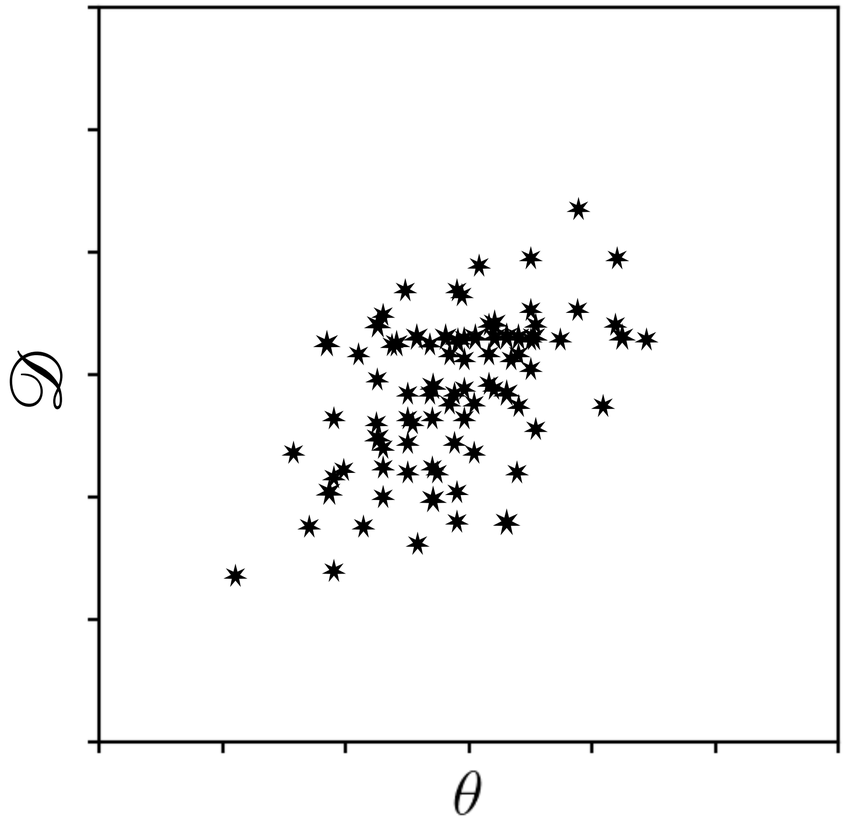
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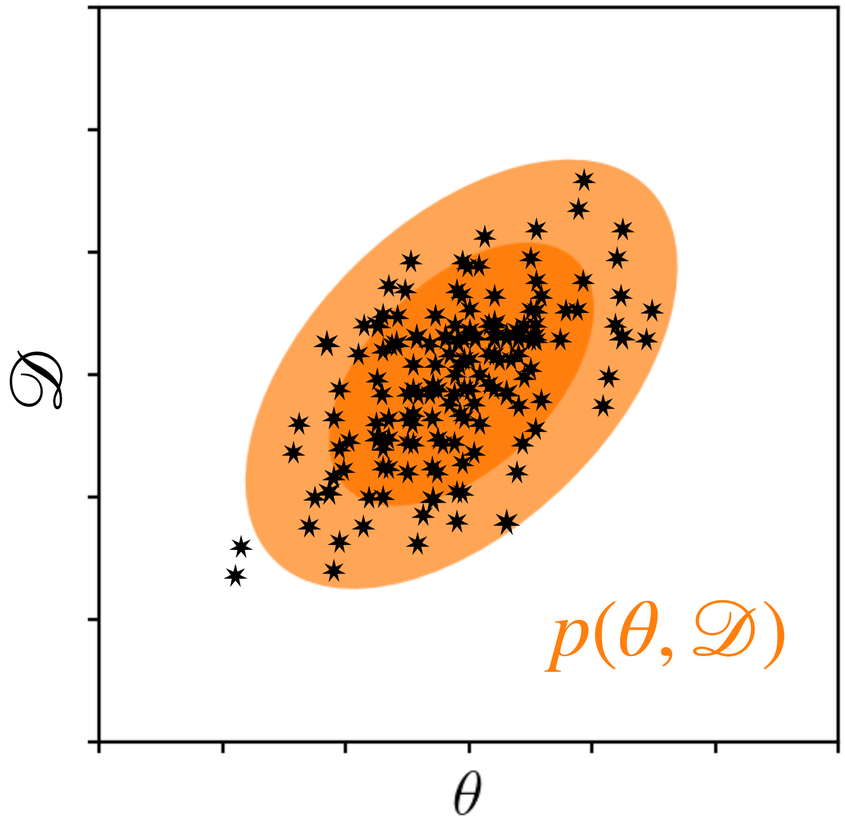
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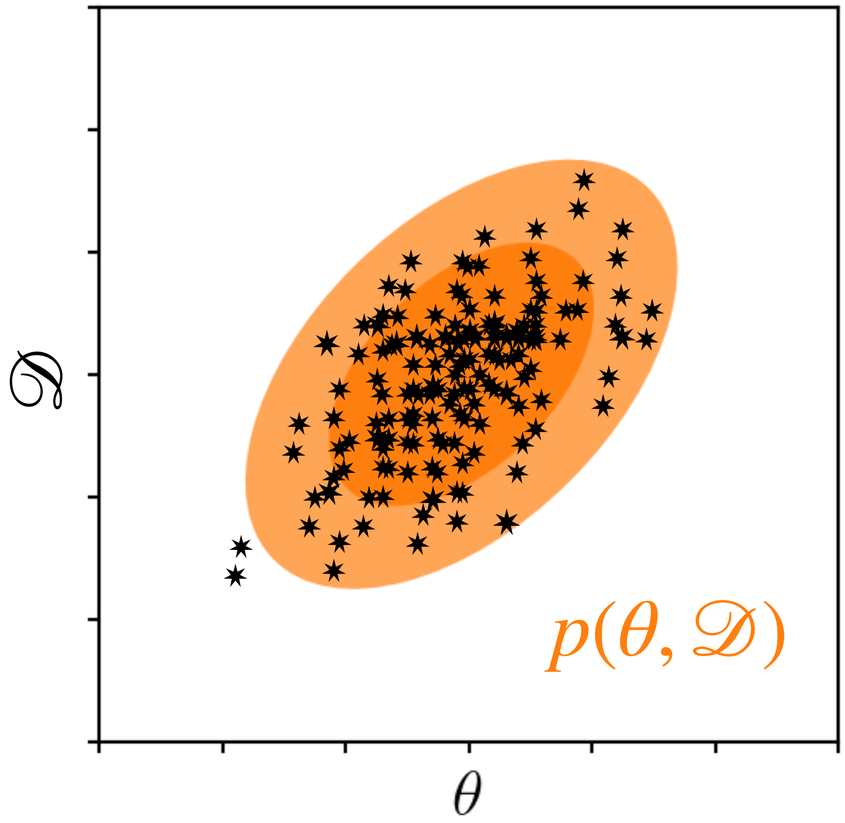


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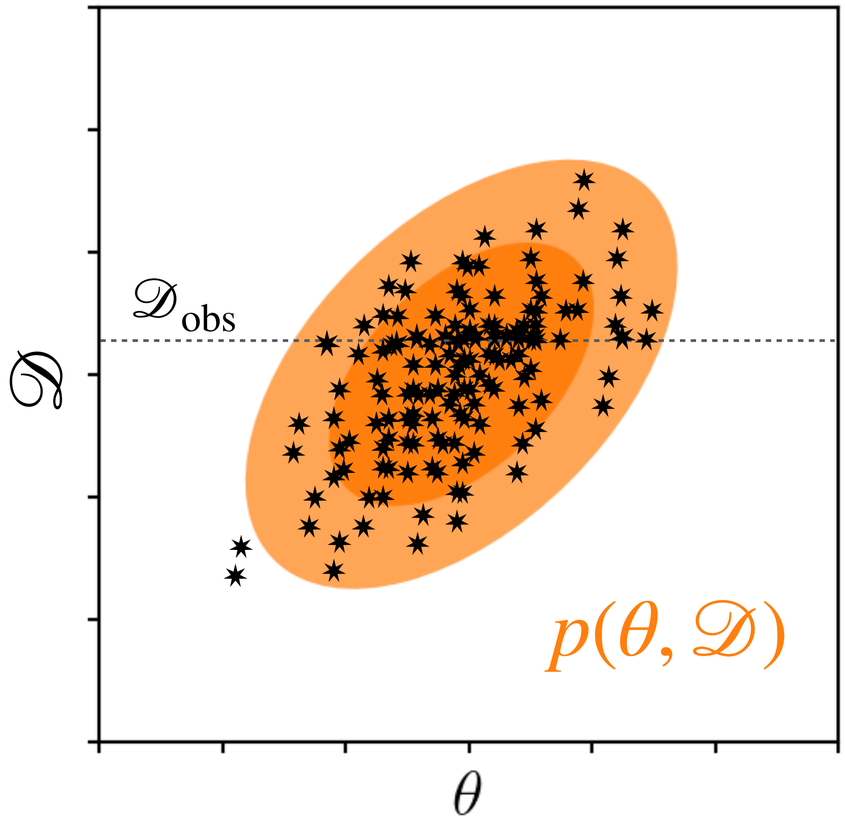


since the simulator samples the likelihood, the simulations sample the **joint distribution** $p(\theta, \mathscr{D})$

ABC in 1-dimensions: posterior for observation $\mathscr{D}_{\text{obs}}$?

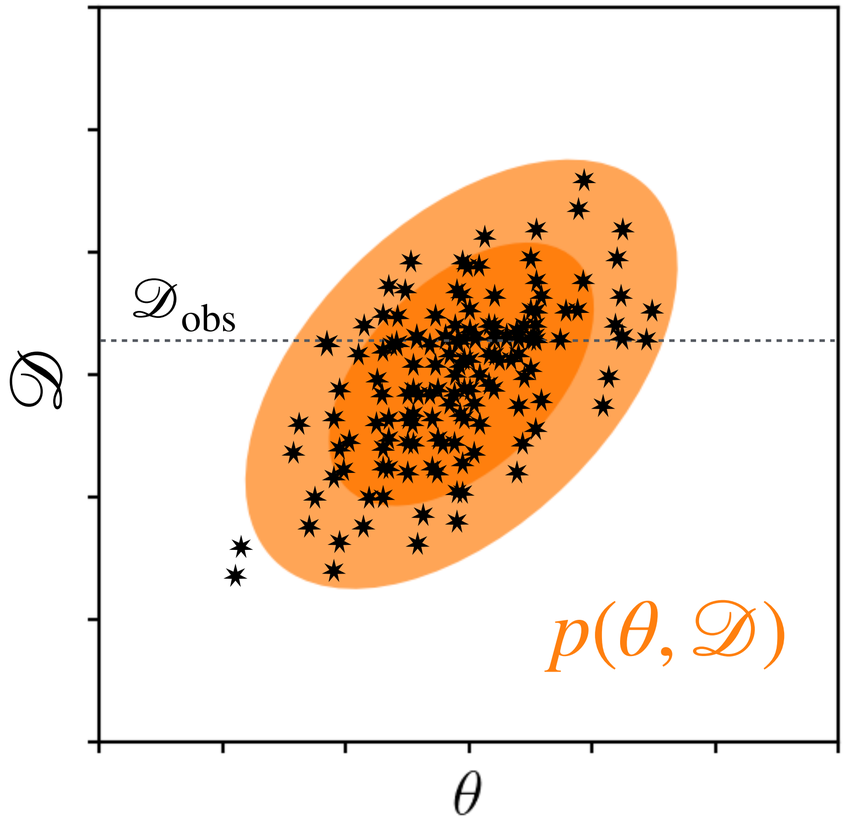


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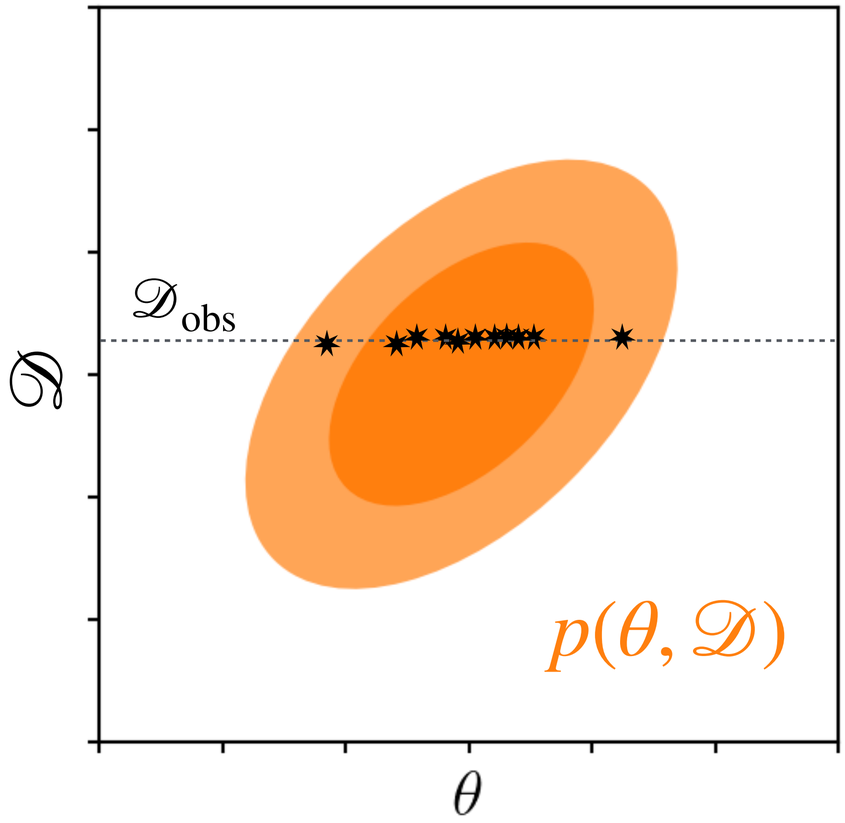
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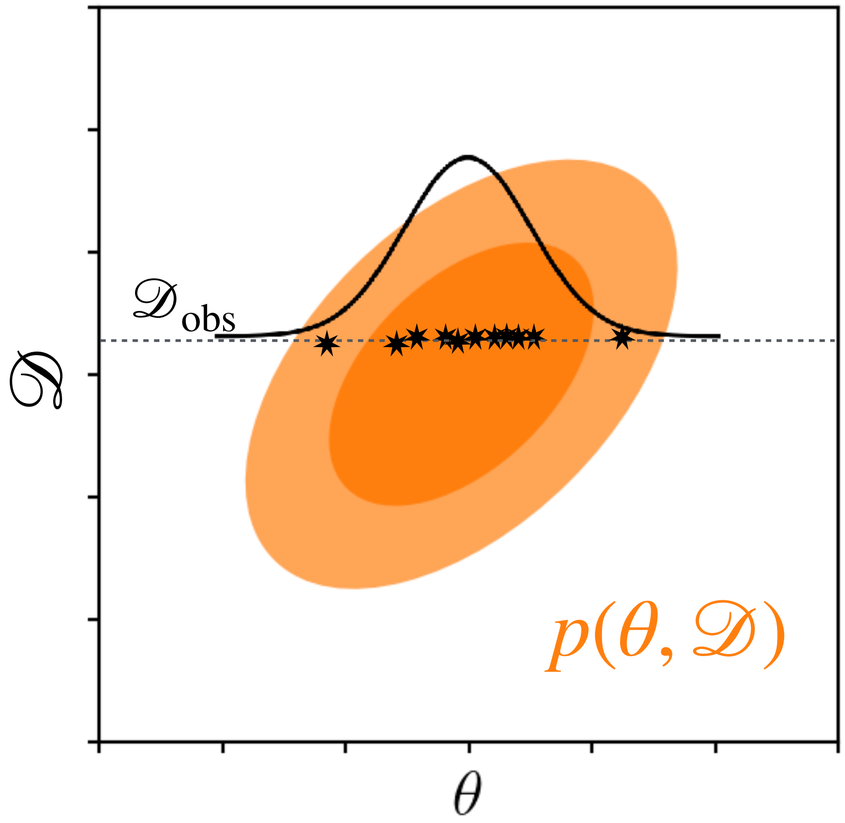
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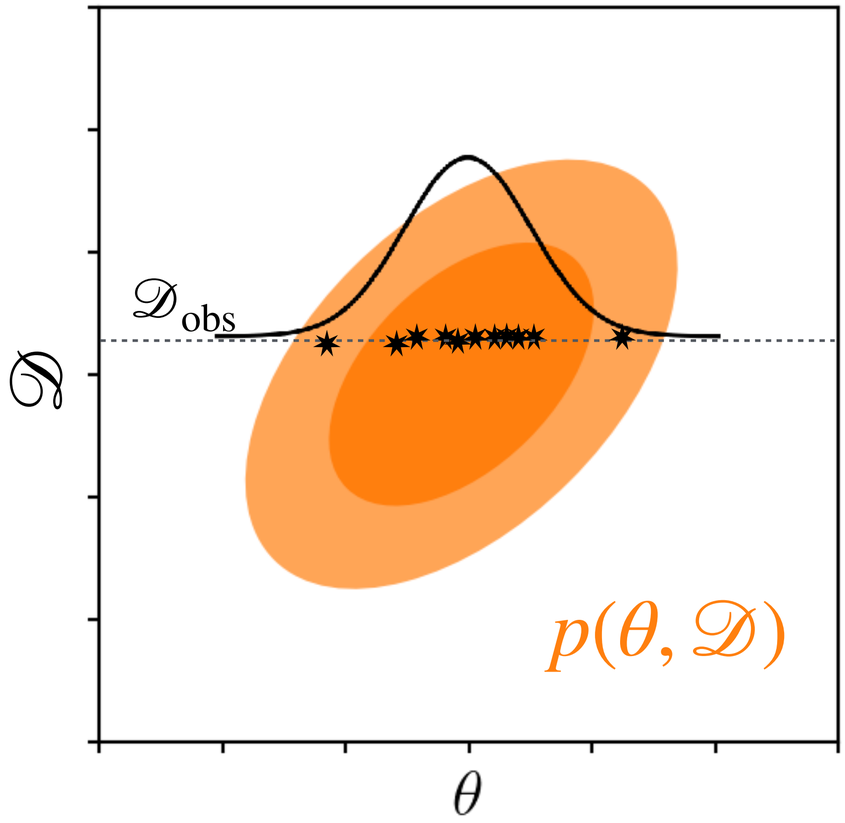


θ distribution of those simulations $= p(\theta \mid \mathscr{D}_{\text{obs}})$

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too strict



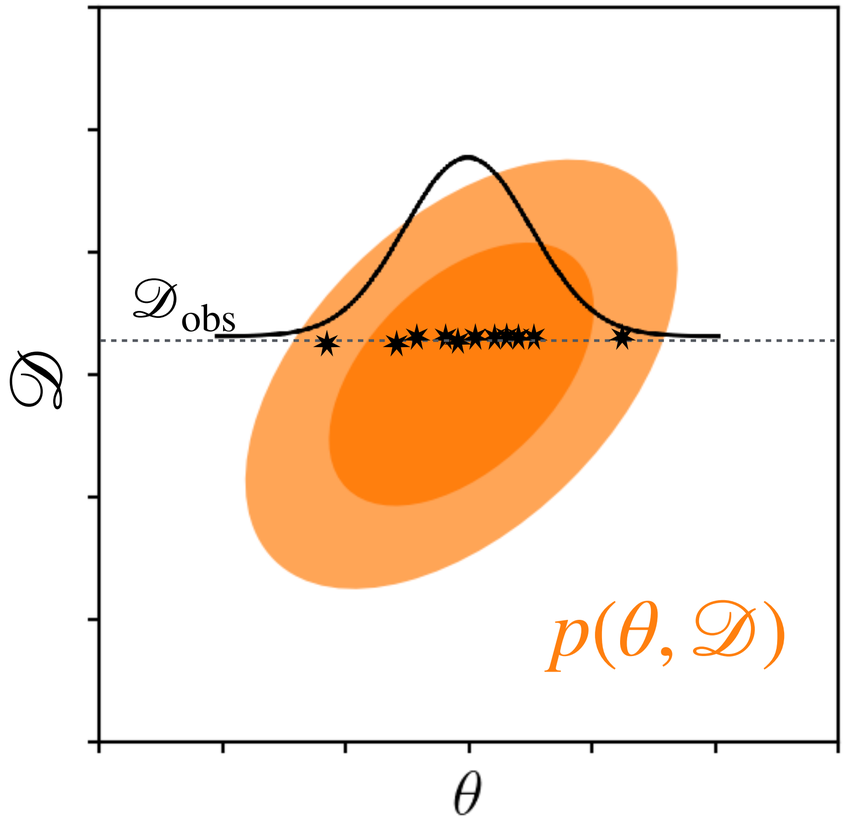
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relaxed to

e.g. $\left|\mathscr{D} - \mathscr{D}_{\text{obs}}\right| < \epsilon$



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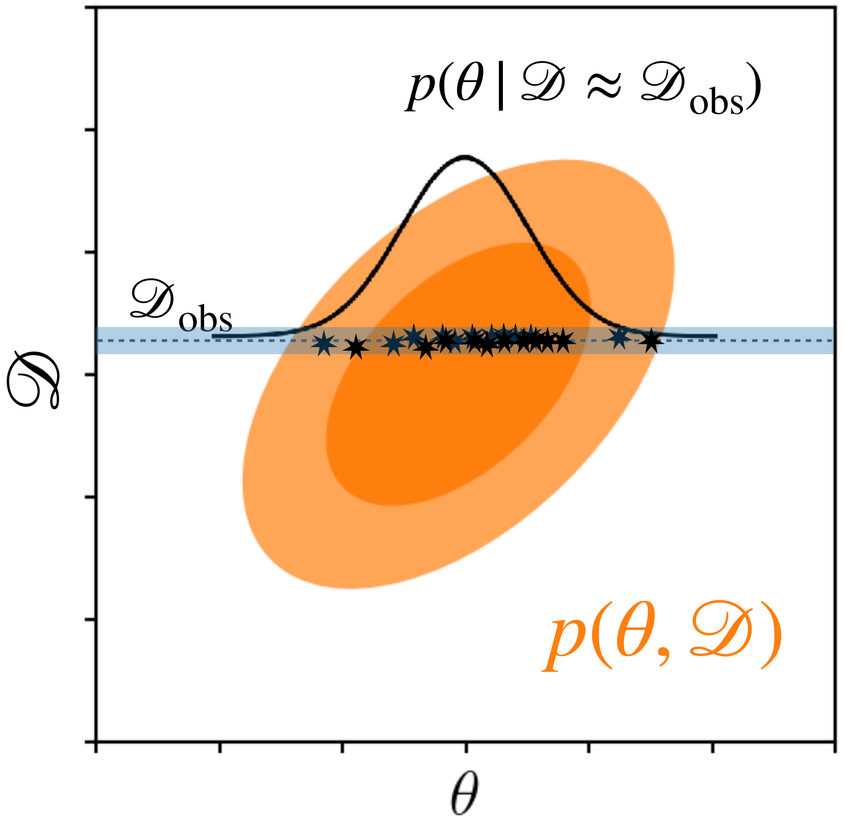
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ABC in practice — higher dimensions

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difficult to define in higher dimensions

$$\approx p(\theta \mid \rho(\mathscr{D}_{\text{obs}}, \mathscr{D}) < \epsilon)$$

define some distance metric ρ arbitrarily small threshold ϵ

groups:

Group 1: Ines, Jayden, Asa
Group 2: Madeline, Alexander, Noah
Group 3: Nikhil, Jules, Michelle, Kaleo
Group 4: Tae Bong, Nicholas, Olivia
Group 5: Bella, Bocheng, Shreya, John

in-class: ABC demo and in-class exercise

ast384c/inclass/lec19_abc/abc_exercise.ipynb